OCEANSCOPE INDIA

High-Performance 4D Digital Twin & 3D Underwater Visualization Engine of the Indian Ocean

Platform: OceanScope India (Smart India Hackathon 2026 · PS SIH_2026)
Domain: Full Indian Ocean Basin (20.0°E → 125.0°E, -45.0°S → 32.0°N)
Core Technologies: React 19, Three.js, React Three Fiber, Custom GLSL Shaders, FastAPI, SQLite, ETOPO Bedrock Bathymetry, Argovis in-situ CTD profiles, 25-Year 4D Multimodal Cubes

1. Executive Summary & System Overview

OceanScope India is a high-performance 4D Digital Twin and 3D Underwater Visualization Engine built to deliver real-time exploration, analysis, and simulation across the Indian Ocean basin. Spanning 25 years of multimodal physical, chemical, and biological oceanographic data (2000–2024+), OceanScope integrates satellite remote sensing, global reanalysis models, in-situ profiling instruments (Argo floats, gliders, moored buoys), and high-resolution bedrock bathymetry into a unified, photorealistic WebGL environment.

The system provides dual interactive operational modes:
1. **4D Interactive Earth Globe:** Full-disk planetary navigation with 25-year timeline scrubbing, dynamic shader layers (Thermal SST, Salinity, Chlorophyll-a, Velocity Current Flow, Seafloor Relief), regional 3D bounding box selection, and sub-grid telemetry inspection.
2. **Immersive 3D Underwater Dive Explorer:** First-person 6-DOF submarine dive mode rendering 3D bedrock bathymetry meshes, volumetric light caustics, realistic underwater fog extinction, and animated marine fauna ecosystems.

2. Geographic Domain & 4D Data Specifications

Domain Parameter	Specification	Operational Function
Geographic Region	Indian Ocean (20.0°E → 125.0°E, -45.0°S → 32.0°N)	Covers Arabian Sea, Bay of Bengal, Andaman Sea, Equatorial Currents, Southern Ocean boundary
Horizontal Grid	309 Latitudes × 421 Longitudes (0.25° Resolution)	High-density sub-grid mesh enabling real-time bilinear spatial interpolation
Vertical Coverage	16 Discrete Standard Depths (0 to 5,000 meters)	0, 10, 25, 50, 75, 100, 150, 200, 300, 500, 750, 1000, 1500, 2000, 3000, 5000 m
Temporal Horizon	25-Year Historical Atlas (300 Monthly Steps: 2000–2024)	Full multi-decadal retrospective tracking climate variability, IOD, and ENSO teleconnections
Bathymetry Model	NOAA ETOPO 60-arcsecond Bedrock Elevation Grid	True seabed relief with deep oceanic trenches, continental shelves, and seamounts
In-Situ Network	Argovis Float Database & Glider CTD Missions	Over 10,000+ real in-situ profiles with pressure, temperature, and salinity calibration

3. Multimodal Oceanographic Visualization Layers

Layer Name	Data Palette / Units	Physical Oceanographic Phenomenon Rendered
1. Natural Earth	Photorealistic True Color	NASA Blue Marble surface land topography, bathymetric shading, cloud cover, and vector coastlines
2. Thermal (SST)	Turbo / Magma (2°C → 32°C)	Sea Surface Temperature, deep thermocline gradients, coastal upwelling (Somali & Malabar), and marine heatwaves
3. Salinity (SSS)	Viridis / Haline (30 → 38 PSU)	Arabian Sea high-salinity evaporative waters vs Bay of Bengal low-salinity river discharge plumes (Ganges-Brahmaputra)
4. Chlorophyll-a	Emerald / Bio (0 → 3.0 mg/m³)	Biological productivity, phytoplankton blooms, nutrient feeding frenzies, and coastal fisheries
5. Ocean Currents	Geodesic Vector Particles (0 → 2 m/s)	Animated vector streamlines rendering the Somali Jet, equatorial Wyrtki jets, gyres, and East India Coastal Currents
6. Ocean Bed (Relief)	Bathymetric Navy (-11,000m → 0m)	Seafloor elevation, Sunda/Java subduction trench, Chagos-Laccadive ridge, Mumbai continental shelf

4. Key Interactive Capabilities & Engines

• **Sub-Grid Telemetry Inspector:** Hover or click anywhere on the ocean globe to sample latitude, longitude, seabed depth, surface temperature, salinity, and real-time current velocity with bilinear interpolation.

• **Vertical CTD Profiler:** Inspects full vertical water column cross-sections down to 5,000 meters with automated thermocline and mixed-layer detection.
• **Regional 3D Bounding Box Selector:** Interactive spatial boundary cropping tool allowing users to isolate marine regions (e.g. Mumbai Shelf, Sunda Trench, Gulf of Aden) and dive directly into 3D space.
• **Subsurface 3D Dive Simulation (WASD/6-DOF):** Pilot an underwater submarine with depth throttle, dynamic water caustics, benthic fog extinction, and marine life (whales, dolphins, manta rays, fish schools).
• **Historic Extreme Events Simulator:** Replays the 2004 Sumatra Tsunami hourly shockwave propagation across the Indian Ocean basin with wave height and arrival time physics.
• **AI Oceanographic Assistant ('Poseidon AI'):** Natural-language marine intelligence interface powered by LLM query synthesis for maritime research and operational queries.

5. Technical Architecture & Performance Pipeline

Frontend: React 19, TypeScript, Vite, Three.js, React Three Fiber, Custom GLSL Shaders, Lucide Icons, Vanilla CSS.
Backend: FastAPI, Uvicorn, SQLite database (oceanscope.db), NumPy, SciPy.
Binary Multidimensional Data Vault: 4D binary data cubes (data/processed/cubes/*.bin) and ETOPO terrain grids (data/processed/terrain/*.bin).
High-Speed Local Data Engine: Self-contained mathematical synthesis generates all 25 years of 4D ocean cubes locally in ~25 seconds on CPU with 0 cloud/download dependencies.

6. Platform Visualizations & Operational Modes

The following figures showcase live interactive rendering modes of the OceanScope India visualization engine:



Figure 1: Full-disk interactive 3D Earth globe with 25-year timeline navigation, telemetry inspector, and sub-grid coordinate sampling.

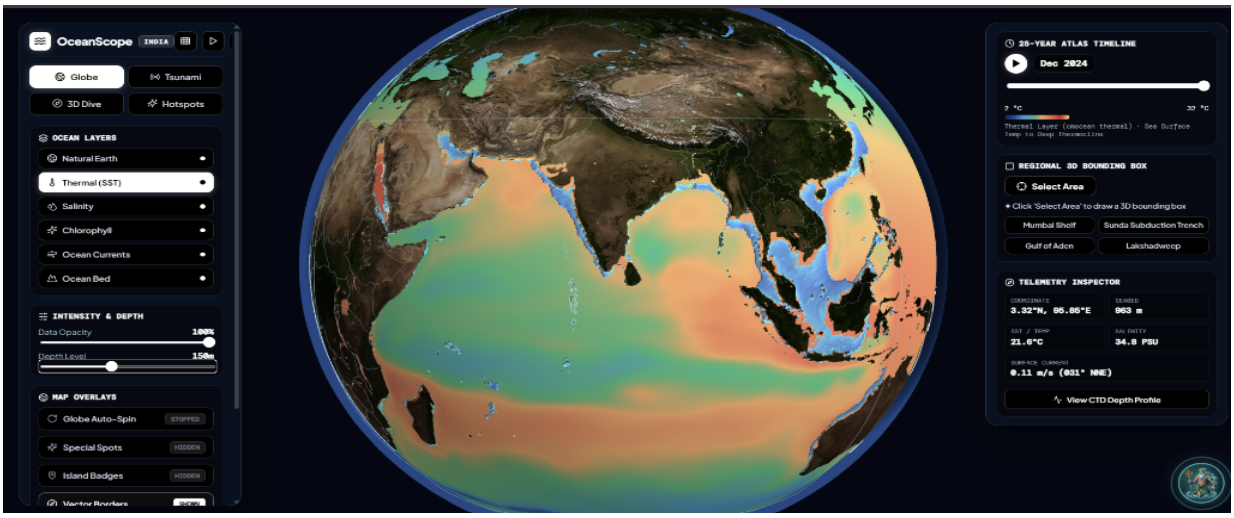


Figure 2: Thermal Layer (SST & Subsurface Thermocline) displaying coastal upwelling along the western coast of India, the Somali Current, and deep thermocline gradients.

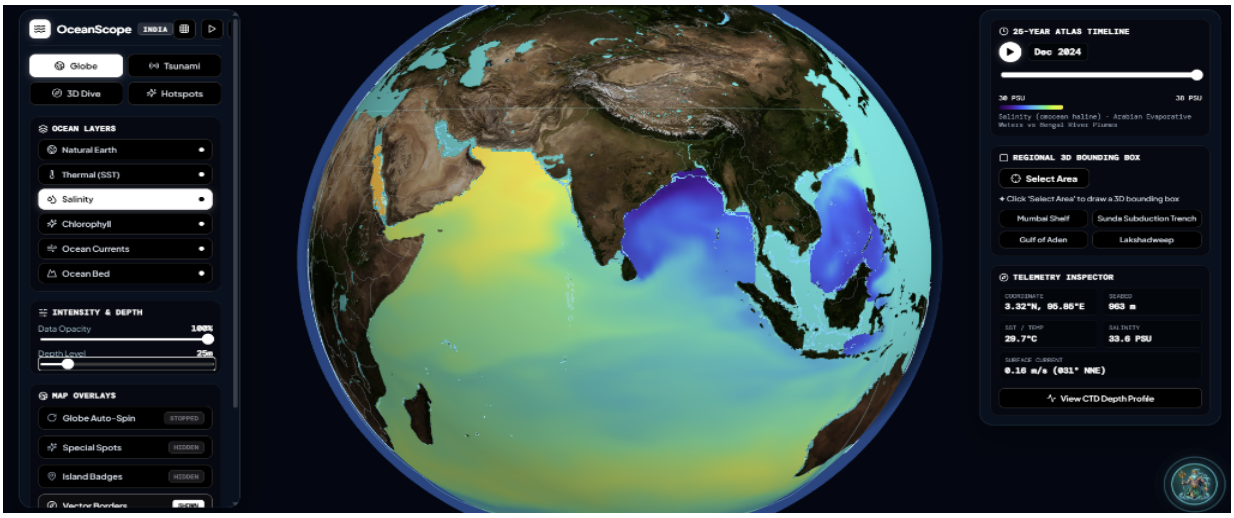


Figure 3: Salinity Layer showcasing high-salinity evaporative waters in the Arabian Sea versus low-salinity freshwater river discharge plumes in the Bay of Bengal.

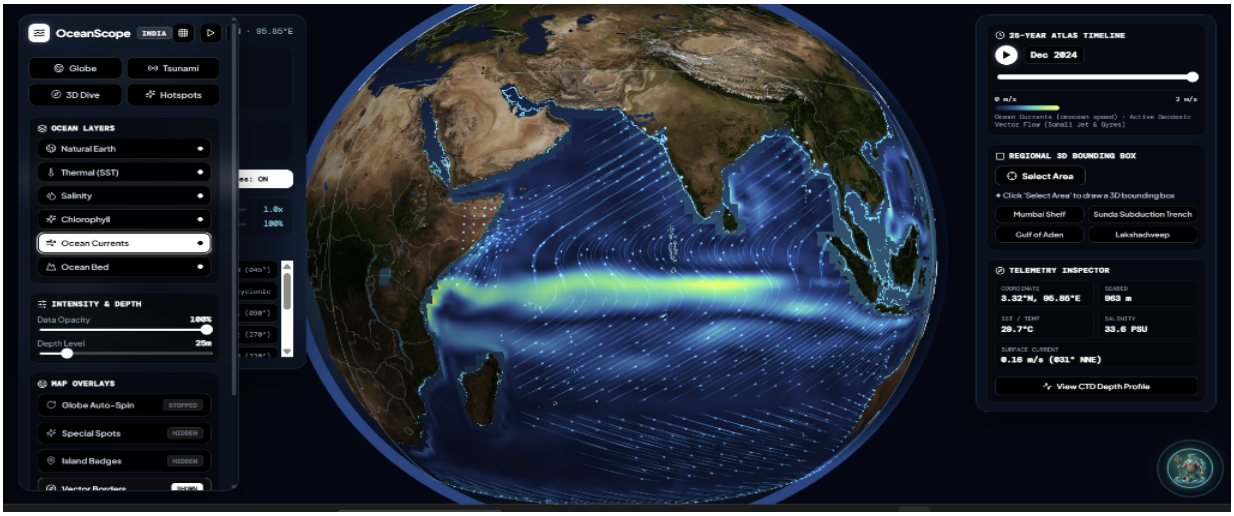


Figure 4: Geodesic Surface Current Streamlines and velocity vector field illustrating equatorial jets, Somali gyres, and East India Coastal Currents (EICC).

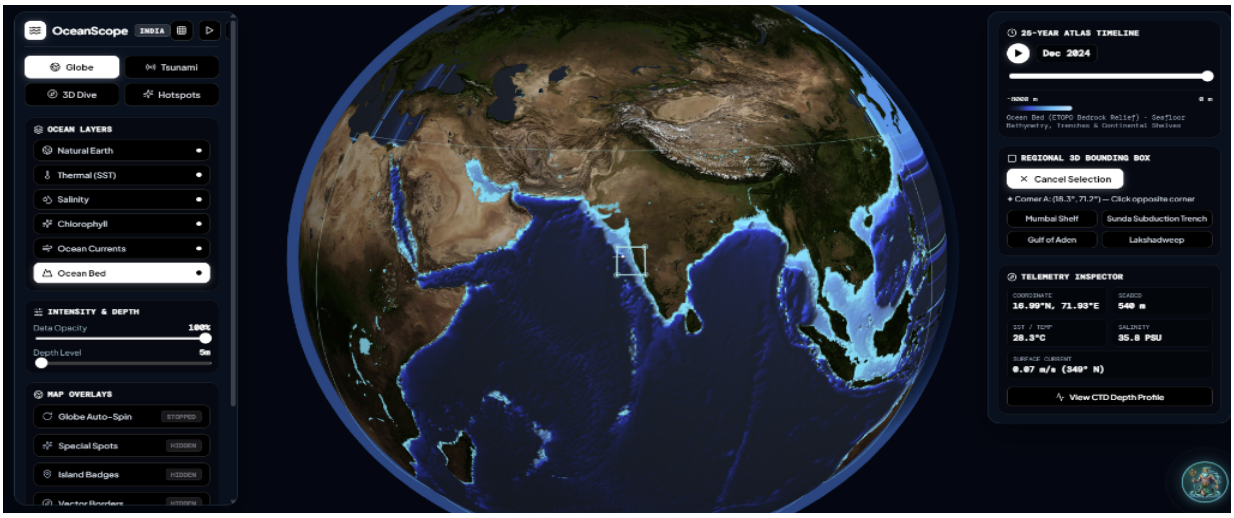


Figure 5: Dual-scale seafloor bedrock bathymetry (ETOPO) showing oceanic trenches, continental shelves, and the interactive 3D bounding box selector.

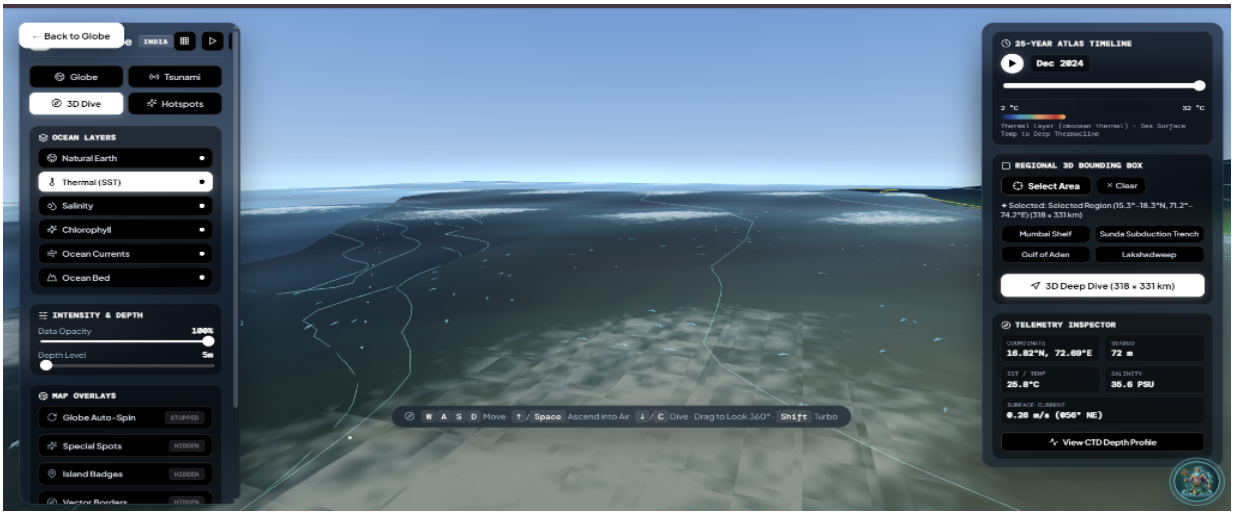


Figure 6: Real-time 3D underwater dive simulation rendering seabed bathymetry mesh, benthic terrain elevation, and underwater optical fog extinction.

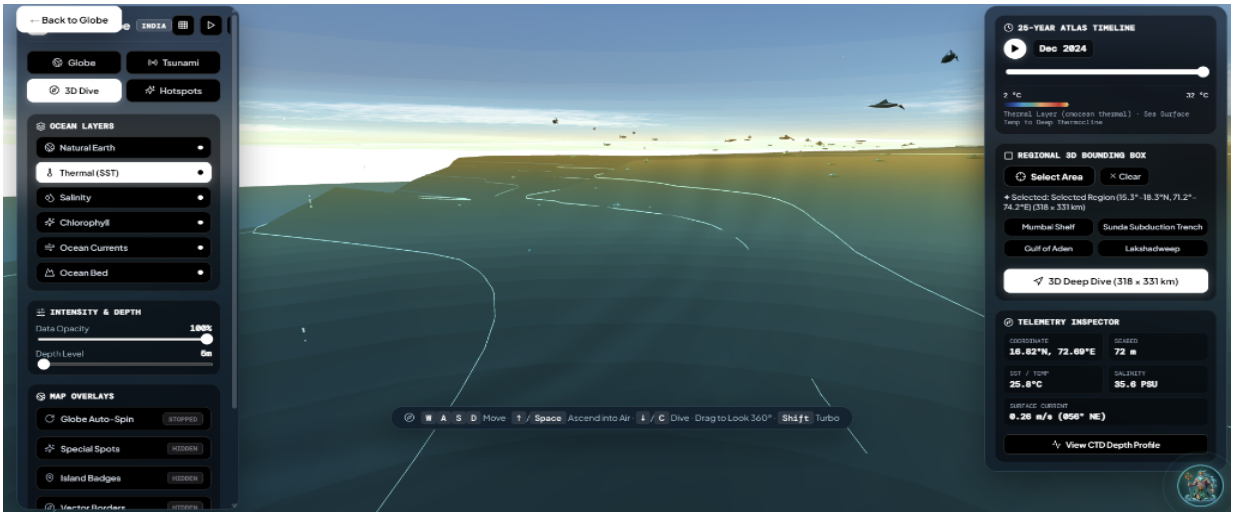


Figure 7: Subsurface 3D perspective along the Mumbai continental shelf showcasing bathymetric escarpments, marine life fauna, and water column stratification.

7. Strategic Value for INCOIS, MoES & Maritime Operations

Strategic Pillar	Operational Deliverable & Impact
1. 4D Ocean Intelligence	Delivers unified 4D situational awareness of the Indian Ocean across 25 years with instant depth-rail navigation and sub-grid telemetry.
2. Subsurface 3D Exploration	Bridges the gap between 2D satellite maps and true 3D underwater bathymetric topography via real-time 6-DOF submarine dive simulation.
3. Disaster & Hazard Preparedness	Interactive physics simulation of historic extreme events, including the 2004 Sumatra Tsunami and marine heatwave anomalies.
4. Fisheries & Ecosystem Monitoring	High-resolution chlorophyll-a and coastal upwelling tracking for marine fishery feeding grounds along Somali and Malabar coasts.
5. Local-First Resilient Architecture	Zero-cloud dependency with local generation of 25-year 4D cubes in ~25s on CPU, ensuring uninterrupted operation for security and defense.

8. Quickstart & Deployment Guidelines

1. Launch Backend Service: python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8000 --reload
• REST API: http://127.0.0.1:8000/ | Health: http://127.0.0.1:8000/health
2. Launch Frontend Application: npm run dev (WebGL UI: http://localhost:5173/)
3. Instant 25-Year Data Sync: npm run sync:data (or python backend/scripts/sync_pipeline.py)